

Sharp affine Hyers–Ulam constants for ℓ_∞ -products

A substantial partial result for arXiv:math/0201098

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Status: substantial partial result, likely valid. Exact sharpness propagates through arbitrary max products. This settles many new non-Hilbert cases but not the source question for every Banach space.

1 Source question and notation

A map $f : E \rightarrow E$ is an ε -*nearisometry* if

$$| \|f(x) - f(y)\| - \|x - y\| | \leq \varepsilon \quad (x, y \in E).$$

Let $H(E)$ be the infimum of the constants c such that every surjective ε -nearisometry of the real Banach space E is within $c\varepsilon$ of a surjective affine isometry. Let $J(E)$ be the Jung constant: the infimum of r such that every subset of E of diameter at most 2 is contained in a ball of radius r .

Väisälä's survey [1] recalls the general upper bound $H(E) \leq J(E)$ and asks on printed page 4 whether it is sharp for every Banach space.

It is natural to ask whether the bound $d(S, f) \leq J(E)\varepsilon + t$ is sharp for every Banach space E . An affirmative answer for Hilbert spaces was given in [HV] but the general case is open.

One might also think that the result holds with better bounds if only continuous maps $f : E \rightarrow F$ are considered. However, this is not the case at least for Hilbert spaces, as proved in [HV].

Figure 1: The exact sharpness question in [1, p. 4].

Huuskonen and Väisälä [2] prove $H(E) = J(E)$ for every Hilbert space and record the same equality for c , c_0 , spaces with $J(E) = 1$, and one regular-hexagon plane. The following theorem propagates any such equality through max products.

2 Idea of the proof

The max norm decouples both sides of the problem. Coordinate projections turn a diameter-two set in a product into diameter-two sets in every factor; coordinatewise Jung centers then assemble

into one bounded product center. For the lower Hyers–Ulam bound, insert a bad nearisometry in one coordinate and leave all other coordinates fixed. The max operation preserves the same additive error.

The only possible complication is that an ambient isometry might mix coordinates. It cannot do so when it approximates the lifted map: after a standard normalization the lifted map is a bounded distance from the identity, and every affine isometry a bounded distance from the identity has identity linear part. Thus any approximant is merely a translation, whose bad coordinate recovers the original factor obstruction.

3 Product theorem

For a nonempty family $(X_i)_{i \in I}$ of nonzero real Banach spaces, write

$$X = \left(\prod_{i \in I} X_i \right)_{\infty} = \left\{ (x_i) : \sup_i \|x_i\| < \infty \right\}, \quad \|(x_i)\| = \sup_i \|x_i\|.$$

Theorem 1 (max-product propagation). *For $X = (\prod_i X_i)_{\infty}$,*

$$J(X) = \sup_{i \in I} J(X_i) \quad \text{and} \quad H(X) \geq \sup_{i \in I} H(X_i).$$

Consequently, if

$$\sup_i H(X_i) = \sup_i J(X_i),$$

then $H(X) = J(X)$. In particular this conclusion holds whenever $H(X_i) = J(X_i)$ for every i .

Proof. Set $J_* := \sup_i J(X_i)$. Each X_i embeds isometrically as a coordinate subspace of X , and projecting a covering ball in X onto that coordinate shows $J(X) \geq J(X_i)$. Hence $J(X) \geq J_*$.

Conversely, let $A \subset X$ have diameter at most 2, fix $a^0 \in A$, and let $\eta > 0$. For every i , the coordinate set $A_i = \{a_i : a \in A\}$ has diameter at most 2, so there is $c_i \in X_i$ with

$$\sup_{a \in A} \|a_i - c_i\| \leq J(X_i) + \eta \leq J_* + \eta.$$

Moreover,

$$\|c_i\| \leq \|c_i - a_i^0\| + \|a_i^0\| \leq J_* + \eta + \|a^0\|,$$

uniformly in i . Thus $c = (c_i)$ belongs to X and $A \subset B_X(c, J_* + \eta)$. Letting $\eta \downarrow 0$ proves $J(X) = J_*$.

We next prove $H(X) \geq H(X_k)$ for a fixed $k \in I$. Let $c \in H(X_k)$. By the definition of $H(X_k)$ there are $\varepsilon > 0$ and a surjective ε -nearisometry $f : X_k \rightarrow X_k$ whose distance from every surjective affine isometry exceeds $c\varepsilon$.

Translate f so that $f(0) = 0$. The fundamental nearisometry theorem [1, Theorem 2.2] supplies a surjective linear isometry $U : X_k \rightarrow X_k$ with $d(f, U) \leq 2\varepsilon$. Replacing f by $g = U^{-1}f$ preserves all distances from the affine-isometry family and gives

$$d(g, \text{id}_{X_k}) < \infty.$$

Lift g to $G : X \rightarrow X$ by

$$(Gx)_k = g(x_k), \quad (Gx)_i = x_i \quad (i \neq k).$$

This map is surjective. If $a = \|x_k - y_k\|$, $a' = \|g(x_k) - g(y_k)\|$, and $b = \sup_{i \neq k} \|x_i - y_i\|$, then

$$|\|Gx - Gy\| - \|x - y\|| = |\max\{a', b\} - \max\{a, b\}| \leq |a' - a| \leq \varepsilon.$$

Hence G is an ε -nearisometry, and $d(G, \text{id}_X) < \infty$.

Let $T : X \rightarrow X$ be a surjective affine isometry with $d(T, G) < \infty$. By Mazur–Ulam, $Tz = Lz + w$ for a surjective linear isometry L . Since $d(T, \text{id}_X) < \infty$, for every $z \in X$ the vectors

$$t(L - I)z + w = T(tz) - tz$$

remain bounded as $t \rightarrow \infty$. Therefore $L = I$ and T is translation by $w = (w_i)$. Restricting to vectors supported in coordinate k gives

$$d(T, G) \geq \sup_{u \in X_k} \|u + w_k - g(u)\| > c\varepsilon,$$

because $u \mapsto u + w_k$ is an affine isometry of X_k . Thus the lifted map defeats every ambient affine isometry at the same constant, and $H(X) \geq c$. Letting $c \uparrow H(X_k)$ and then taking the supremum over k proves the second assertion.

Finally the general estimate $H(X) \leq J(X)$ from [1], together with the two formulas just proved, gives the equality under the stated hypothesis. $\square$

4 Concrete new cases

Corollary 2. *The equality $H(X) = J(X)$ holds in each of the following cases.*

1. $X = (\prod_i H_i)_\infty$ for arbitrary real Hilbert spaces H_i . In particular,

$$H(X) = J(X) = \sup_i J(H_i).$$

For finite-dimensional H_i of dimensions n_i , the factor constant is $\sqrt{2n_i/(n_i + 1)}$; an infinite-dimensional factor or unbounded finite dimensions give supremum $\sqrt{2}$.

2. $X = c_0 \oplus_\infty Y$ for every real Banach space Y ; here $H(X) = J(X) = 2$.

3. $X = \ell_2^2 \oplus_\infty \mathbb{R}$; here

$$H(X) = J(X) = \frac{2}{\sqrt{3}}.$$

This three-dimensional max product is not a Hilbert space.

Proof. Hilbert factors satisfy $H = J$ by [2]. Also $H(c_0) = J(c_0) = 2$ by [2, Theorem 3.2], while every Banach space has Jung constant at most 2. The final formula uses $J(\ell_2^2) = \sqrt{4/3} = 2/\sqrt{3}$ and $J(\mathbb{R}) = 1$. Apply Theorem 1. $\square$

5 Verification, novelty, and limitations

The proof uses only three externally established facts: Mazur–Ulam, the fundamental 2ε approximation theorem, and the upper bound $H(E) \leq J(E)$. The two product arguments were separately checked for arbitrary (including uncountable) index sets; boundedness of the assembled Jung center is the only point not automatic coordinatewise.

A bounded novelty search through 11 August 2026 checked the run’s exact-id and keyword indexes; exact web searches for “Hyers–Ulam constant direct sum”, “ $H(E) = J(E)$ nearisometry product”, and variants; all nine OpenAlex-indexed papers citing the source survey; and all seven OpenAlex-indexed papers citing [2]. No direct-sum or max-product formula for these constants was located. Novelty confidence is moderate rather than definitive because the topic has sparse and partly pre-arXiv literature.

The full source question remains open. For finite p , the same lifting gives only

$$H(X \oplus_p Y) \geq \max\{H(X), H(Y)\},$$

whereas the Jung constant can be strictly larger: already $J(\mathbb{R} \oplus_2 \mathbb{R}) = 2/\sqrt{3} > 1$. General norm-one complemented subspaces also lack the max metric needed to preserve a uniform additive error. A full solution therefore needs a coupled construction realizing general Jung-extremal configurations, not merely coordinatewise witnesses.

Human-review recommendation. Check especially (i) normalization of the factor nearisometry by a linear isometry, (ii) the assertion that a finite-distance affine isometry has identity linear part, and (iii) boundedness of the product Jung center.

References

- [1] J. Väisälä, *A survey of nearisometries*, in *Papers on Analysis*, Reports of the Department of Mathematics and Statistics, University of Jyväskylä 83 (2001), 305–315; arXiv:math/0201098.
- [2] T. Huuskonen and J. Väisälä, *Hyers–Ulam constants of Hilbert spaces*, *Studia Math.* 153 (2002), 31–40; DOI: 10.4064/sm153-1-3.